Ashwath Sridhar

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EDUCATION

Georgia Institute of Technology - Atlanta, GA **Expected December 2028**
Bachelor of Science in Electrical Engineering **GPA: 4.0**

EXPERIENCE

huMannity Medtec – Santa Clarita, CA **June 2026 – August 2026**
Electrical Engineering Intern

- Designed and validated a switching circuit for pulse timing and polarity, identifying and correcting a schematic error incorporated into the official design; achieved 98% timing accuracy and characterized current spikes down to 40 ns
- Automated a 126-point oscilloscope characterization sweep across 18 operating configurations using a custom Python script and GUI, reducing full-sweep test time from an 8-hour manual process to under 3 minutes
- Programmed embedded MCU firmware in Simplicity Studio IDE to validate I2C communication with a power management IC at 1.8V logic, confirming 99.75% measurement accuracy across 3,600 samples at 3 Hz polling spanning a 3.5-4.2V range to support a lower-power design
- Diagnosed and debugged a signal acquisition limitation on an ECG board that failed to detect low-amplitude R-wave signals, then resolved it through amplifier gain optimization to enable reliable peak detection down to 0.5mV

International Services and Programs – Santa Clarita, CA **August 2025 – May 2026**
College Assistant

- Maintained and organized digital and physical records for an international student program serving 800+ students, updating files, correcting outdated information, and keeping documentation structured so staff could quickly locate current materials
- Handled 20+ student inquiries each week related to visa requirements, enrollment, and course scheduling, reviewing documentation and coordinating with staff to provide accurate information before time-sensitive deadlines

Activities

SiliconJackets **August 2026 – Present**
Analog and Mixed Signal Chip Design Member

- Designed a 5-stage current-starved ring VCO with a current-mirror bias network in SKY130 (130 nm CMOS), analytically sizing the transistors from process parameters to meet a 450–550 MHz, under 500 µW target at 1.8 V.
- Debugged frequency drift over 0–70 °C in Cadence Virtuoso simulations by tracing it to PMOS threshold-voltage variation relative to the bias overdrive, and corrected it by lowering the control voltage and lengthening the current-source PFETs, holding oscillation to 451–550 MHz.
- Verified the VCO across transient, temperature, and process-corner (tt, ff, ss) simulations, measuring a 51% duty cycle, 205 ps rise time, and 38–48 µW ring power consumption while driving a 1 pF load through a 6-stage tapered output buffer.
- Completed the full-custom layout of the VCO and buffer with guard rings and multi-finger devices, achieving 0 DRC violations and a clean LVS match in Cadence Pegasus, and presented the results in a documented design review.

HyTech Racing **August 2026 – Present**
Formula SAE Electric Racing Team Member

- Designed, routed, and validated a 2-layer USB-C PCB in Altium Designer, integrating an LM555 astable oscillator and 6-LED array; developed a component library from manufacturer datasheets and achieved a DRC-clean, hardware-verified design
- Developed and tested embedded C++ firmware for an Arduino Uno/BME280 system, implementing I2C and SPI communication and temperature-dependent LED control; verified builds through GitHub CI

FIRST Robotics Team **August 2022 – June 2025**
Lead Programmer

- Led Java development of drivetrain, Limelight vision tracking, and arm control systems, improving autonomous scoring consistency and contributing to a 1st-place finish at SoCal Showdown
- Mentored 10+ programmers in Java and object-oriented development, enabling underclassmen to independently implement and debug robot functionality

SKILLS

Hardware: PCB Design & Layout, Schematic Capture, Circuit Design & Analysis, Analog/Digital Circuits, Board Bring-Up, Hardware Validation, Root-Cause Analysis, Signal Integrity, Embedded Systems, MCU Programming, CMOS Design, DRC/LVS, Soldering
Software & Test Equipment: Altium Designer, Cadence Virtuoso, Cadence Pegasus, LTspice, MATLAB, Simplicity Studio, Git, GitHub CI, Oscilloscope, Function Generator, DC Power Supply, Digital Multimeter, Differential Probe
Programming & Protocols: Python, C++, C, Java, HTML, CSS, I2C, SPI, UART, CAN, USB-C